SSC JE 2026 · ME Paper I · Final read

Read before you leave. No paper is allowed inside.

Thermodynamics ≈14 Q

- Q = ΔU + W; work *by* gas is +. Isothermal: ΔU = 0, Q = W. Throttling: **h const**. Free expansion: Q = W = ΔU = 0.
- Zeroth law → temperature. Kelvin-Planck: no engine turns all heat into work (PMM-2). Clausius: heat won't go cold → hot by itself.
- $\eta_{Carnot} = 1 - T2/T1 \cdot COP_R = T2/(T1-T2) \cdot COP_{HP} = COP_R + 1$. ⚠ Kelvin only. 527/27 °C → 62.5%.
- n = 0 isobaric · 1 isothermal · γ adiabatic · ∞ isochoric. Polytropic W = (p1V1−p2V2)/(n−1); isothermal pV ln(V2/V1).
- Air: R = 0.287 kJ/kg·K, γ = 1.4; cp − cv = R. Open system h = u + pv (pv = flow work).
- Steam: triple pt 0.01 °C; critical 374.15 °C, 221.2 bar; h = hf + x·hfg; hfg @100 °C = 2257 kJ/kg. Mollier = h–s. Throttling calorimeter → x.
- Rankine η ↑ with higher boiler p, superheat, lower condenser p, reheat, regeneration. $\eta_{Brayton} = 1 - 1/rp^{(\gamma-1)/\gamma}$.
- $\oint \delta Q/T \le 0$ (= 0 reversible). Reversible adiabatic = isentropic. Real processes: entropy ↑.
- Steady flow: V2 = √(2Δh) (Δh in J/kg) = 44.72√Δh (kJ/kg).

IC Engines ≈13 Q

- SI: Otto, spark plug, r 6–10, **quantity** governing (throttle). CI: Diesel, injector, r 14–22, **quality** governing. Hit-and-miss: whole cycles skipped.
- 4-stroke: 1 power stroke / 2 rev (n = N/2). 2-stroke: every rev, needs scavenging, mist (petroil) lubrication.
- Octane (iso-octane vs n-heptane) = petrol anti-knock. Cetane (cetane vs α-methylnaphthalene) = diesel ignition quality.
- SI knock at **end** of combustion; CI knock at **start** (ignition delay). What cures one worsens the other.
- Firing order **1-3-4-2**. Morse test → IP of multi-cylinder. Willan's line → FP of CI. Rope brake BP = π(D+d)N(W−S)/60.
- $\eta_{Otto} = 1 - 1/r^{\gamma-1}$ (r 8 → 56.5%). r = (Vs+Vc)/Vc. Same r: Otto > Dual > Diesel. Same p_{max}: Diesel > Dual > Otto.
- IP = p_mLAnk/60, BP = 2πNT/60, η_{mech} = BP/IP, FP = IP − BP. Stirling: 2 isothermal + 2 isochoric (heat rejected at const T).
- Battery ignition: spark voltage falls as speed rises. Magneto: rises with speed.

Boilers · Turbines · Nozzles · Compressors ≈14.5 Q

- Fire-tube: Cochran, Lancashire, Cornish, Locomotive. Water-tube: Babcock & Wilcox, Stirling. High-pressure: Lamont, **Benson (drumless)**, Loeffler, Velox.
- **Mountings** (safety): safety valve, water-level indicator, pressure gauge, fusible plug, stop valve, blow-off cock, feed check valve.
- **Accessories** (efficiency): economiser (heats feed water), **superheater** (steam above saturation), air preheater, feed pump, injector, steam trap (drains condensate).
- Critical pressure ratio: air **0.528** · superheated steam **0.546** · saturated steam **0.577**.
- De Laval = impulse (pressure drops in nozzles only). Parsons = reaction, 50% degree. Compounding cuts rotor speed.
- Compressor: isothermal = least work. Perfect intercooling: p2 = √(p1p3). $\eta_{vol} = 1 + C - C(p2/p1)^{1/n}$, falls as clearance or pressure ratio rises. Surging: centrifugal at low flow.

Refrigeration & AC ≈11 Q

- **1 TR = 3.5 kW = 210 kJ/min**. COP = refrigerating effect / work (5 TR on 5 kW → 3.5).
- VCR: compressor (isentropic) → condenser → expansion valve (**isenthalpic**) → evaporator.
- Halocarbons: R-number + 90 → C,H,F: R-12 CCl_2F_2 · R-22 $CHClF_2$ · R-134a $C_2H_2F_4$. Inorganic R-7xx = 700 + molar mass: NH_3 **R-717**, water R-718, CO_2 **R-744**, SO_2 **R-764**.
- Bell-Coleman = air cycle (aircraft). Absorption: NH_3–water, LiBr–water (water is the refrigerant).
- Hermetic: motor + compressor in one welded shell. Semi-hermetic: bolted cover.
- Saturated air: DBT = WBT = DPT. Sensible heating/cooling: ω constant.

Fluid Mechanics ≈22 Q

- τ = μ du/dy, ν = μ/ρ. 1 poise = 0.1 Pa·s, 1 stoke = 10^{-4} m^2/s. Liquid μ falls with T; gas μ rises.
- p = ρgh (5 m water = 49.05 kPa). Absolute = gauge + atm. Gauge = above atm; vacuum = below. 1 atm = 101.325 kPa = 10.33 m water = 760 mm Hg.
- Piezometer: no gas or negative p. U-tube: yes. Inverted U: small differences. Bourdon: gauge pressure.
- Centre of pressure **below** centroid: $h_{cp} = \bar{x} + I_G/(A\,\bar{x})$; does not depend on liquid density. Floating body stable if GM > 0.
- Δp: drop 4σ/d · bubble 8σ/d · jet 2σ/d. Capillary h = 4σcosθ/ρgd; mercury falls.
- A1V1 = A2V2 (half d → 4 × V). Bernoulli: steady, incompressible, non-viscous, along a streamline. TEL is $V^2/2g$ above HGL.
- Venturi Cd 0.95–0.98 · orifice 0.6–0.65 · pitot V = √(2gh). Hg–water manometer h = 12.6x.
- **Re < 2000 laminar, > 4000 turbulent.** Laminar pipe: f = 64/Re, $V_{max} = 2V_{avg}$, velocity parabolic, shear stress **linear** (0 at centre).
- $h_f = fLV^2/2gD$. Losses: sudden expansion $(V1-V2)^2/2g$ · exit $V^2/2g$ · entry $0.5V^2/2g$. Max power when $h_f = H/3$.
- **No-slip:** any real (viscous) fluid has zero velocity relative to a fixed wall.
- Rectangular notch $Q \propto H^{3/2}$; V-notch $Q \propto H^{5/2}$.

Hydraulic Machines ≈8 Q

- **Pelton**: impulse, high head, low Ns, tangential; best bucket speed u = V/2. **Francis**: reaction, medium head, radial/mixed. **Kaplan**: reaction, low head, high discharge, axial, adjustable blades. Draft tube: reaction only.
- Ns: turbine $NvP/H^{5/4}$ · pump $NvQ/H^{3/4}$.
- Centrifugal pump: priming, backward-curved vanes, NPSHa > NPSHr (no cavitation). Series → heads add; parallel → discharges add. $Q \propto N$, $H \propto N^2$, $P \propto N^3$. Double volute balances radial thrust.
- Reciprocating pump: air vessel smooths flow; negative slip = actual > theoretical.

Production ≈6 · 12–14 in 2024

- Allowances: shrinkage, draft, machining, distortion, **shake (only negative)**. Riser freezes last; Chvorinov $t \propto (V/A)^2$.
- Investment = lost wax (wax pattern + ceramic shell). Die / permanent mould = metal mould, non-ferrous. Centrifugal = pipes.
- Defects: blow holes (gas) · cold shut, misrun (low fluidity) · hot tear (restrained contraction).
- Flame: neutral O_2:C_2H_2 = 1:1, ≈3200 °C · oxidising → brass, bronze · carburising → high-C steel.
- TIG: non-consumable W. MIG: consumable wire. Arc welding: electrode is consumed. Spot/seam = resistance. Thermit = rails.
- Brazing > 450 °C, soldering below; filler drawn in by **capillary action**; base metal not melted.
- V = πDN/1000 m/min (D 25 mm, V 15.7 → 200 rpm). Taylor $VT^n = C$: n HSS 0.1 · carbide 0.25 · ceramic 0.5.
- Chips: continuous (ductile, high speed) · discontinuous (brittle) · BUE (ductile, low speed). Merchant ϕ = 45° + α/2 − β/2.
- Grinding: Al_2O_3 → steel; SiC → cast iron, non-ferrous. Order: abrasive, grain, grade, structure, bond.
- EDM: conductive + dielectric · ECM: Faraday · USM: hard brittle · EBM: vacuum · LBM: any. Down = climb milling. Shaper: quick return.
- Punching: punch = hole size, clearance on die. Blanking: die = blank size, clearance on punch. F = L·t·τ. Hot work above recrystallisation.

Materials · SOM · EM · ToM ≈11 Q

- Fe–C: eutectoid **0.8% C, 727 °C** (pearlite); eutectic **4.3% C, 1147 °C** (ledeburite). Steel ≤ 2% C; low-C < 0.3%.
- Anneal (furnace cool) softens · normalise (air) refines · harden (quench) → martensite · temper cuts brittleness.
- BCC (α-Fe, Cr, W, Mo) APF 0.68, 2 atoms · FCC (γ-Fe, Al, Cu, Ni) 0.74, 4 · HCP (Zn, Mg, Ti) 0.74, 6.
- Brass Cu+Zn · bronze Cu+Sn · Invar Fe+36% Ni · SS 18-8 (Cr-Ni) · HSS 18-4-1 (W-Cr-V).
- E = 2G(1+μ) = 3K(1−2μ), μ ≤ 0.5. Sudden load: 2P/A. Thermal σ = EαΔT (free bar: 0).
- BM max: SSB centre WL/4 · SSB UDL $wL^2/8$ · cantilever WL, $wL^2/2$. δ: $WL^3/48EI$ · $5wL^4/384EI$ · $WL^3/3EI$ · $wL^4/8EI$.
- T/J = τ/r = Gθ/L, J = $\pi d^4/32$, torsional rigidity GJ, P = 2πNT/60. Euler Le: H-H L · F-F L/2 · F-free 2L · F-H L/√2. Hoop pd/2t = 2 × long.
- Angle of repose = friction angle, tan φ = μ. Range max at 45°. Sphere CG at its centre.
- Belt $T1/T2 = e^{\mu\theta}$; Tc = Tmax/3. Grashof s + l ≤ p + q. Whitworth = single slider-crank. Stiffer Hartnell spring → less sensitive. Isochronous = zero speed range.
- Failure: ductile → von Mises (best) or Tresca; brittle → max principal stress (Rankine). Soderberg most conservative.

Calculator and steam table are on screen (calculator icon left of your photo; steam table under "Useful Data"). Skip anything that needs a table lookup until round 4.

Reasoning · GA · Current affairs · Exam day

Exam day

- Take: printed admit card (**write nothing on it** before the exam), **original** photo ID with the **same DOB**, transparent water bottle.
- Not allowed: any watch, phone, earphones, calculator, pen, paper. No jewellery. No mask. No toilet break.
- Inside: write the certification statement, sign, left thumb impression, mother's name. Find the calculator icon and "Useful Data" steam table. Get the invigilator to sign before leaving.
- Qualify: OBC 25% = **50/200**. Target **120+ net** for Paper II shortlisting.

120-minute plan

1	GA	10 min	Only instant answers, ≤ 15 s each
2	Mechanical	55 min	Theory + one-step; mark anything > 90 s
3	Reasoning	35 min	Series, coding, ranking, clock first; puzzles last
4	Marked	20 min	ME numericals, then puzzles

- −0.25 per wrong. Blind guess +0.06 · one option out +0.17 · two out +0.38. **Guess only after ruling one out**; ≤ 10 blind guesses.
- Units first (K, N-mm-MPa, radians). Options 10 × apart → units error; 2 × or 4 × → ratio error. Re-read: radius or diameter? gauge or absolute?
- Last 2 min: nothing "marked" without the answer you mean.

Reasoning shortcuts

- Series: differences → 2nd differences → ×/÷ → squares/cubes ± → alternate terms. 2, 6, 12, 20, 30, **42**.
- A=1 … Z=26; EJOTY = 5,10,15,20,25. Opposite letter = 27 − n.
- Coding: find the shift (+1, −1, reverse, opposite). CAT → DBU so DOG → **EPH**.
- Symbol swap: rewrite first, then BODMAS.
- Ranking: total = left + right − 1 (12th + 18th → 29).
- Direction: draw it; distance $\sqrt{(x^2 + y^2)}$. Morning shadow → west; evening → east.
- Blood relation: draw the tree.
- Clock: `θ = |30H − 5.5M|` (3:30 → 75°). Hands overlap 22 times a day.
- Calendar: normal year +1 day, leap +2. Century leap only if ÷ 400.
- Syllogism: smallest Venn. "All A are B" → "Some B are A" always.
- Mirror: left ↔ right. Water: top ↔ bottom. Standard die: opposite faces sum to 7.
- Missing number: row sum → row product → (a × b) + c.

Science

- Units: pressure Pa · energy J · power W · charge C · flux Wb · flux density T · capacitance F · inductance H · luminous intensity cd.
- Light 3×10^8 m/s; sound ≈ 343 m/s in air, fastest in solids, none in vacuum. Escape velocity 11.2 km/s.
- Myopia → concave lens; hypermetropia → convex. Blue sky: scattering. Diamond sparkle: TIR.
- 1 kWh = 3.6×10^6 J · 1 HP = 746 W · −40 °C = −40 °F.
- Hygrometer humidity · lactometer milk · hydrometer density · anemometer wind · sphygmomanometer BP · pyrometer high T.
- Acids: vinegar acetic · lemon citric · curd lactic · ant formic · tamarind tartaric · tomato oxalic.
- Baking soda $NaHCO_3$ · washing soda $Na_2CO_3{\cdot}10H_2O$ · bleaching powder $CaOCl_2$ · PoP $CaSO_4{\cdot}½H_2O$ · gypsum $CaSO_4{\cdot}2H_2O$ · laughing gas N_2O.
- Air 78% N_2, 21% O_2. Crust O > Si > Al. Liquid metal Hg; liquid non-metal Br. Galvanising = Zn.
- Ores: bauxite Al · haematite Fe · galena Pb · cinnabar Hg.
- Vitamins: A night blindness · B1 beriberi · B3 pellagra · B12 pernicious anaemia · C scurvy · D rickets · K clotting.
- O universal donor, AB universal recipient. RBC life 120 days. Largest gland liver; smallest bone stapes; 46 chromosomes.
- Malaria: Plasmodium via female Anopheles. Viral: AIDS, polio, measles, dengue, rabies. Bacterial: TB, typhoid, cholera, tetanus.

Polity · History · Geography · Economy

- Constitution adopted 26 Nov 1949, in force 26 Jan 1950. 42nd Amd (1976): socialist, secular, integrity; Fundamental Duties (51A).
- FR Part III, Art 12–35: 14 equality · 17 untouchability · 19 freedoms · 21 life · 21A education (86th) · 32 remedies. DPSP Part IV (Ireland).
- President 52 · impeachment 61 · VP 63 (RS chairman) · PM 75 · AG 76 · money bill 110 · budget 112 · SC 124 · CAG 148 · EC 324.
- Emergency: national 352 · President's rule 356 · financial 360. Amendment 368.
- Panipat 1526, 1556, 1761 · Plassey 1757 · Buxar 1764 · Revolt 1857 Meerut · INC 1885 (Hume; first president W.C. Bonnerjee).
- Bengal partition 1905 · Jallianwala 13 Apr 1919 · Non-Cooperation 1920 · Chauri Chaura 1922 · Dandi 1930 · Quit India 1942.
- IST 82.5° E (Mirzapur). Largest state Rajasthan; most populous UP; smallest Goa. Tropic of Cancer: 8 states. Palk Strait: India–Sri Lanka.
- Corbett first NP (1936) · Kaziranga rhino · Gir lion · Sundarbans tiger.
- RBI 1935, nationalised 1949. CPI target 4% ± 2%. NITI Aayog 1 Jan 2015. GST 1 Jul 2017. SEBI statutory 1992. NABARD 1982.

Current affairs 2026

- Men's T20 WC: **India** beat NZ (Ahmedabad, 8 Mar), 3rd title. PoT Sanju Samson; final Bumrah.
- IPL: **RCB** (2nd in a row). Orange Cap Vaibhav Sooryavanshi (15, youngest); Purple Cap Rabada.
- Women's T20 WC: Australia (beat England, Lord's), 7th title.
- FIFA WC: **Spain** 1-0 Argentina (aet). Golden Ball Rodri, Boot Mbappé, Glove Unai Simón.
- Wimbledon: Sinner; Linda Nosková (first major).
- CWG Glasgow: India 4th, 39 medals (13 gold); Mirabai Chanu 3rd straight gold.
- Asian Games Aichi-Nagoya: India 4th, 85 medals (21 gold); men's hockey gold (5-1 Malaysia).
- Chess Olympiad Samarkand: India Open silver, women's bronze.
- Nobel: Peace Navi Pillay · Literature Anne Carson · Chemistry Kagan & Soai · Medicine Deisseroth, Hegemann, Nagel (optogenetics) · Economics Hall & Woodford.
- Intl Booker: *Taiwan Travelogue* (Yáng Shuāng-zǐ, tr. Lin King).
- Magsaysay: Tommy Koh, Runa Khan, Bo Kyi. Film awards: *Article 370*; Mammootty & Kartik Aaryan; Yami Gautam.
- Army chief Gen **Dhiraj Seth** · Air chief ACM **Ashutosh Dixit** (29th) · CJI **Surya Kant** (53rd) · RBI **Sanjay Malhotra** · NITI CEO Anurag Jain.
- Repo **5.50%** (+25 bps, Oct). Q1 FY27 GDP 7.8%. DA 63%.
- NSAP → PM Samajik Sahayata Yojana. EOS-05 (GSLV-F17): first imaging satellite for GEO. Gaganyaan G1 (Vyommitra) pending.
- BRICS chair India 2026 (summit New Delhi). SCO summit Bishkek.